

Predicting neuroblastoma clinical outcomes with gene expression profiles: A comparative analysis of models and platforms

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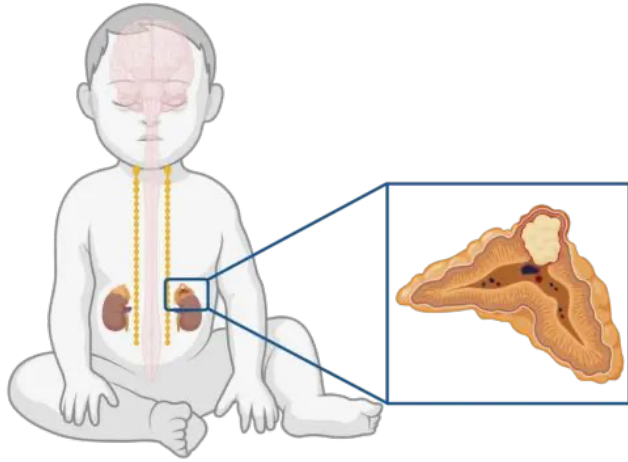


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Introduction

Neuroblastoma



- pediatric cancer
- develops from tissues that form the sympathetic nervous system
- majority of cases are non familiar
- high risk neuroblastoma usually has MYCN amplification

Causes not well understood

Goal

Build a lean model to predict death from disease based on microarray and RNA-Seq data.

Dataset

498 gene expression profiles

- microarray
- RNA-seq

Randomly divided in training and test dataset

Primary endpoint:

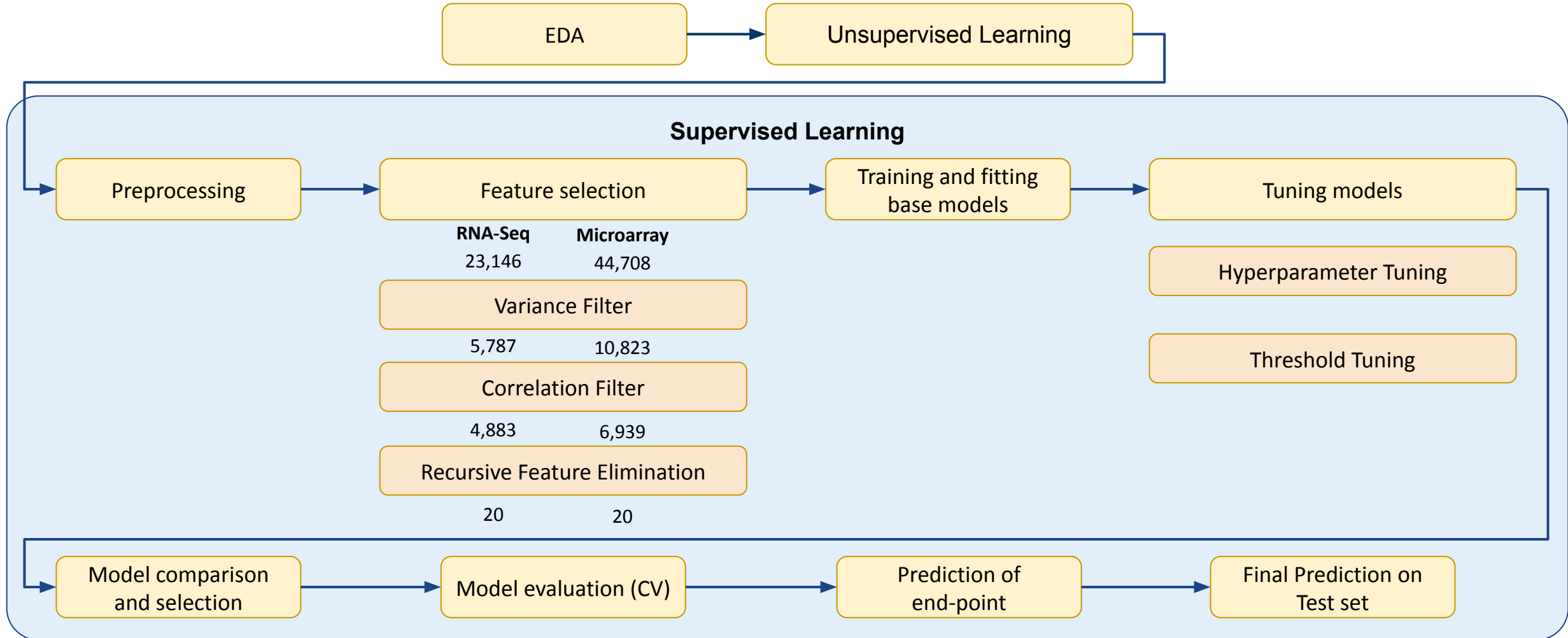
- death from disease

Secondary endpoints:

- high risk
- progression
- International Neuroblastoma Staging System (INSS)

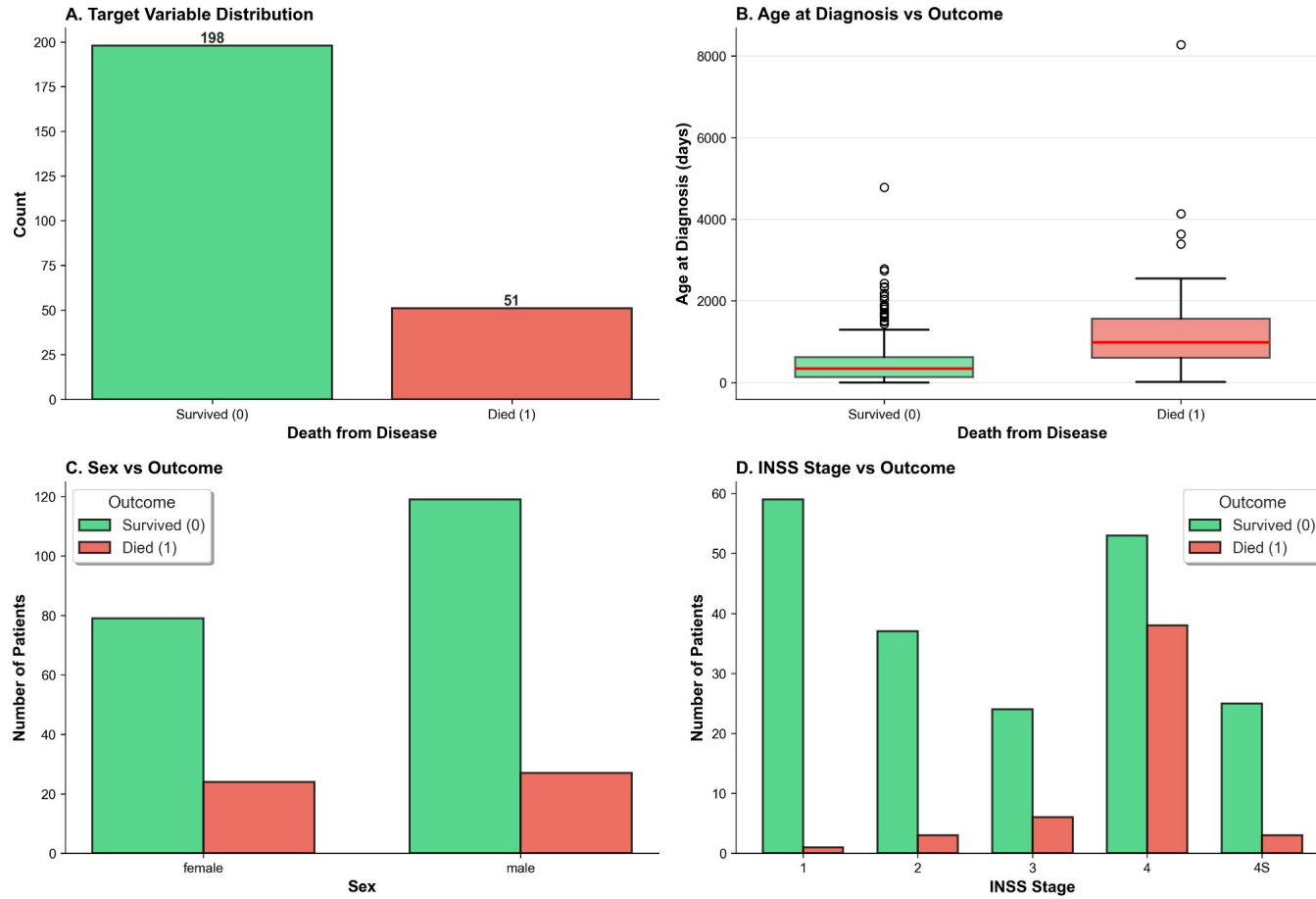


Analysis Flow Chart

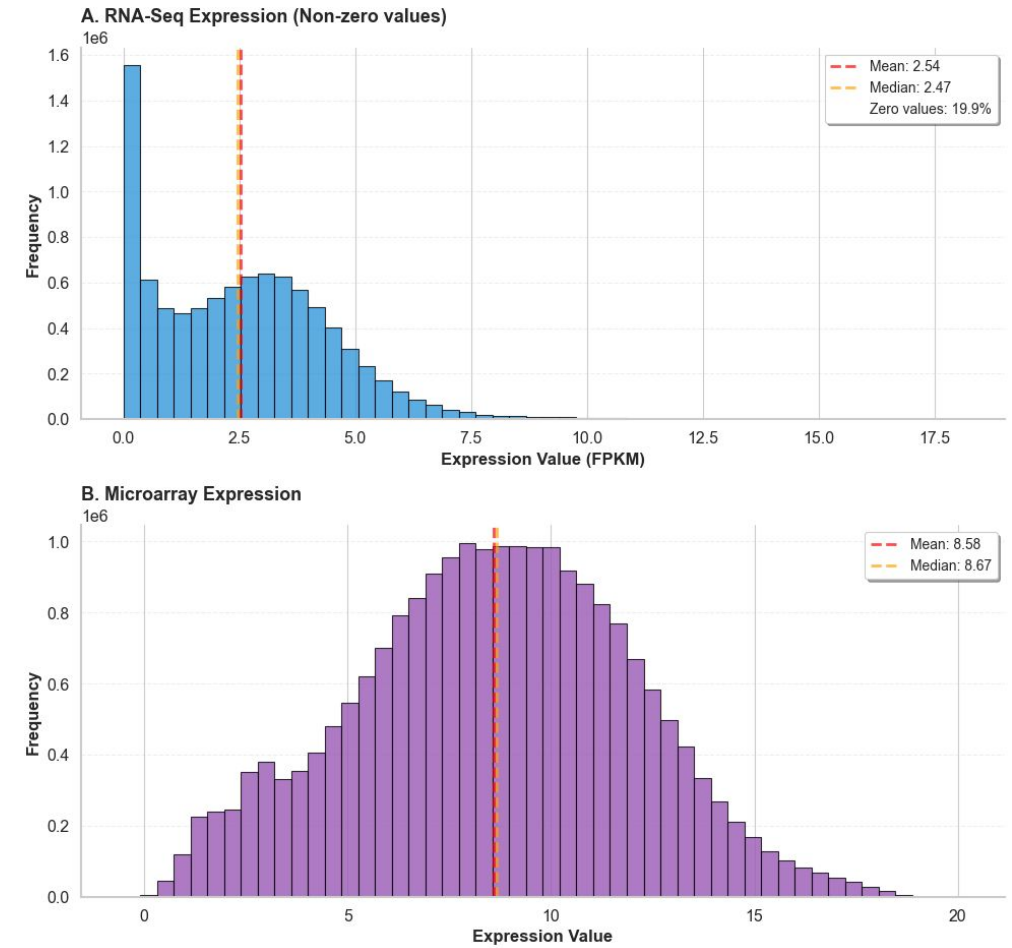


EDA

Clinical Features and Outcome Analysis



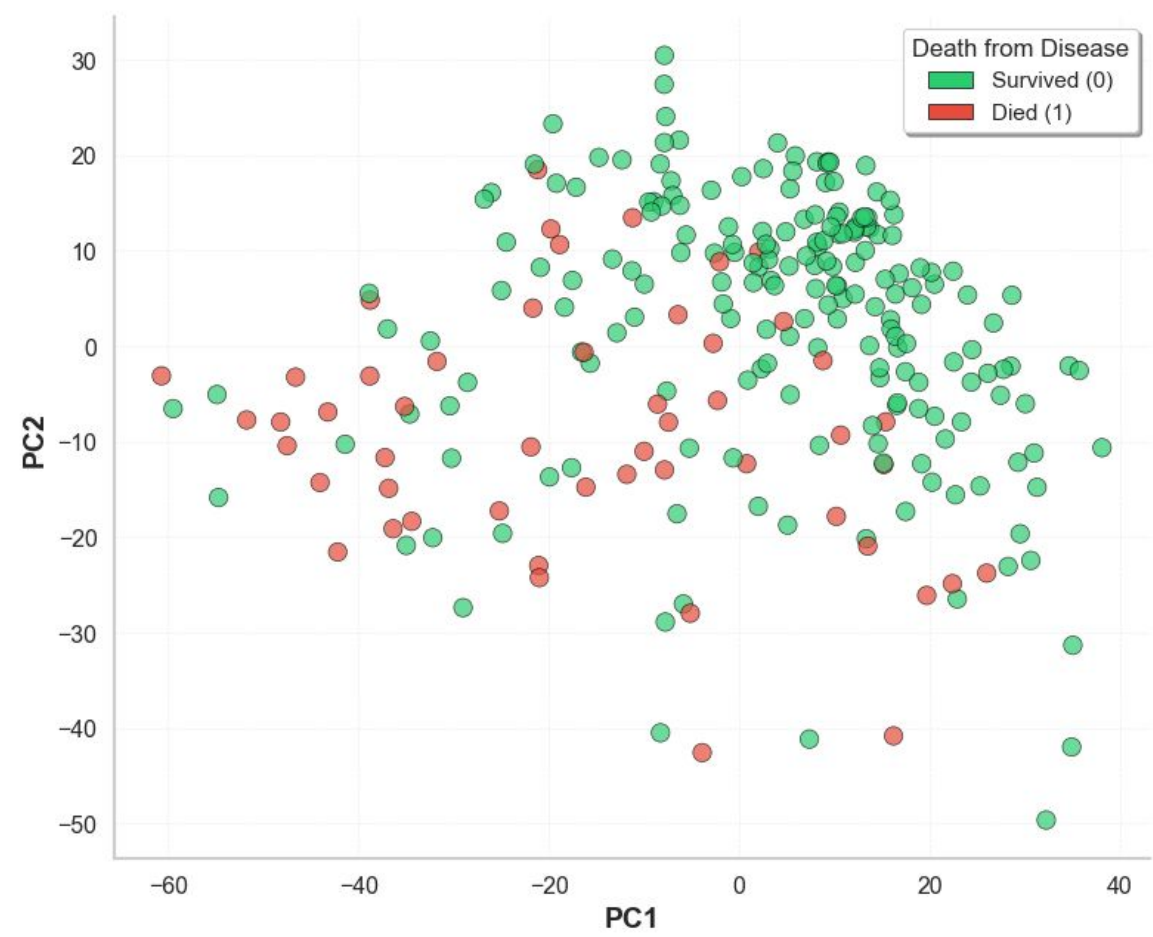
Global Expression Distributions



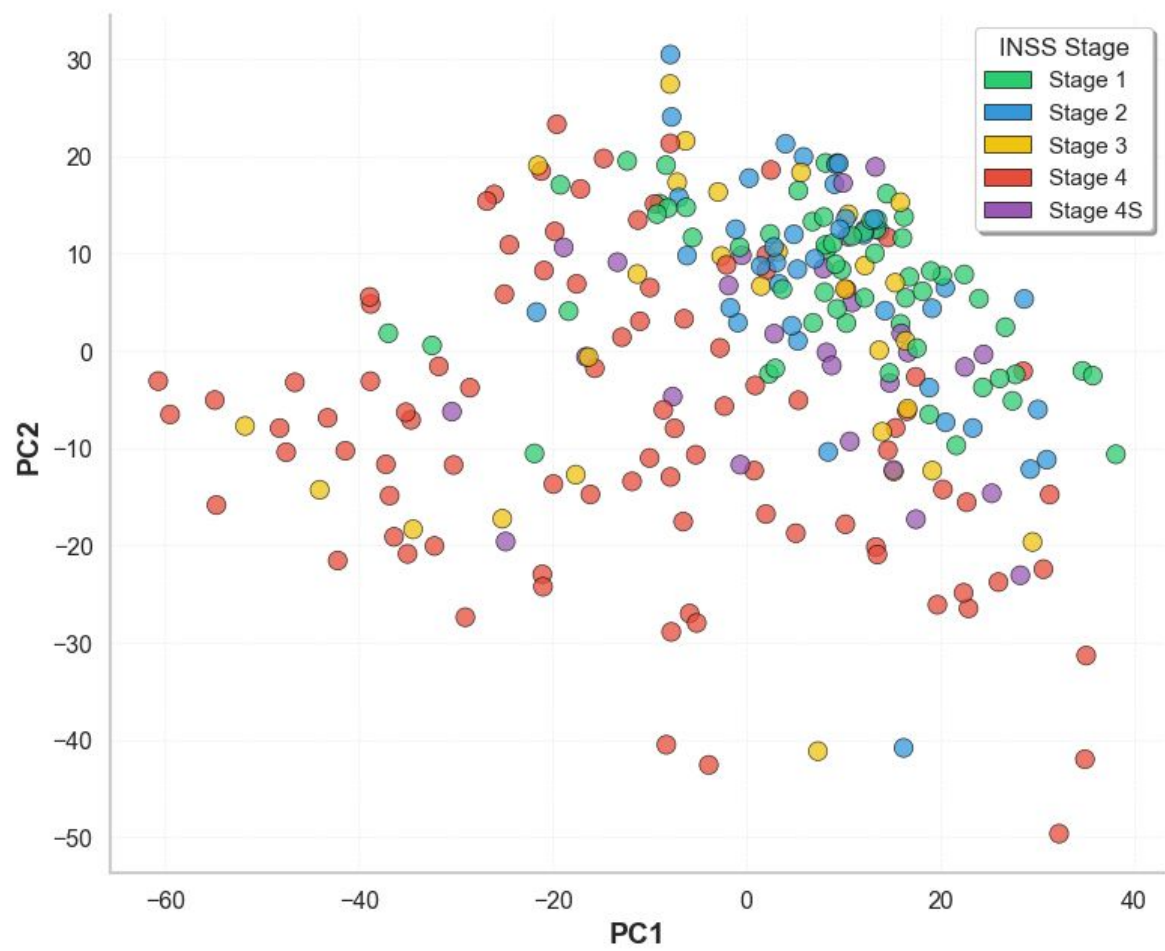
Unsupervised Machine Learning

RNA-Seq PCA Indicates Limited Separability of Clinical and Stage Labels

PCA of RNA-Seq Expression (Top 2000 Variable Genes)



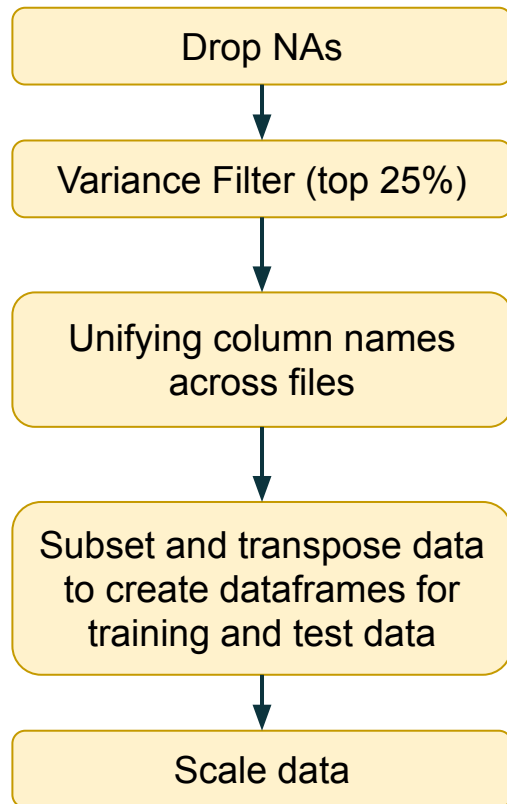
PCA of RNA-Seq Expression Colored by INSS Stage



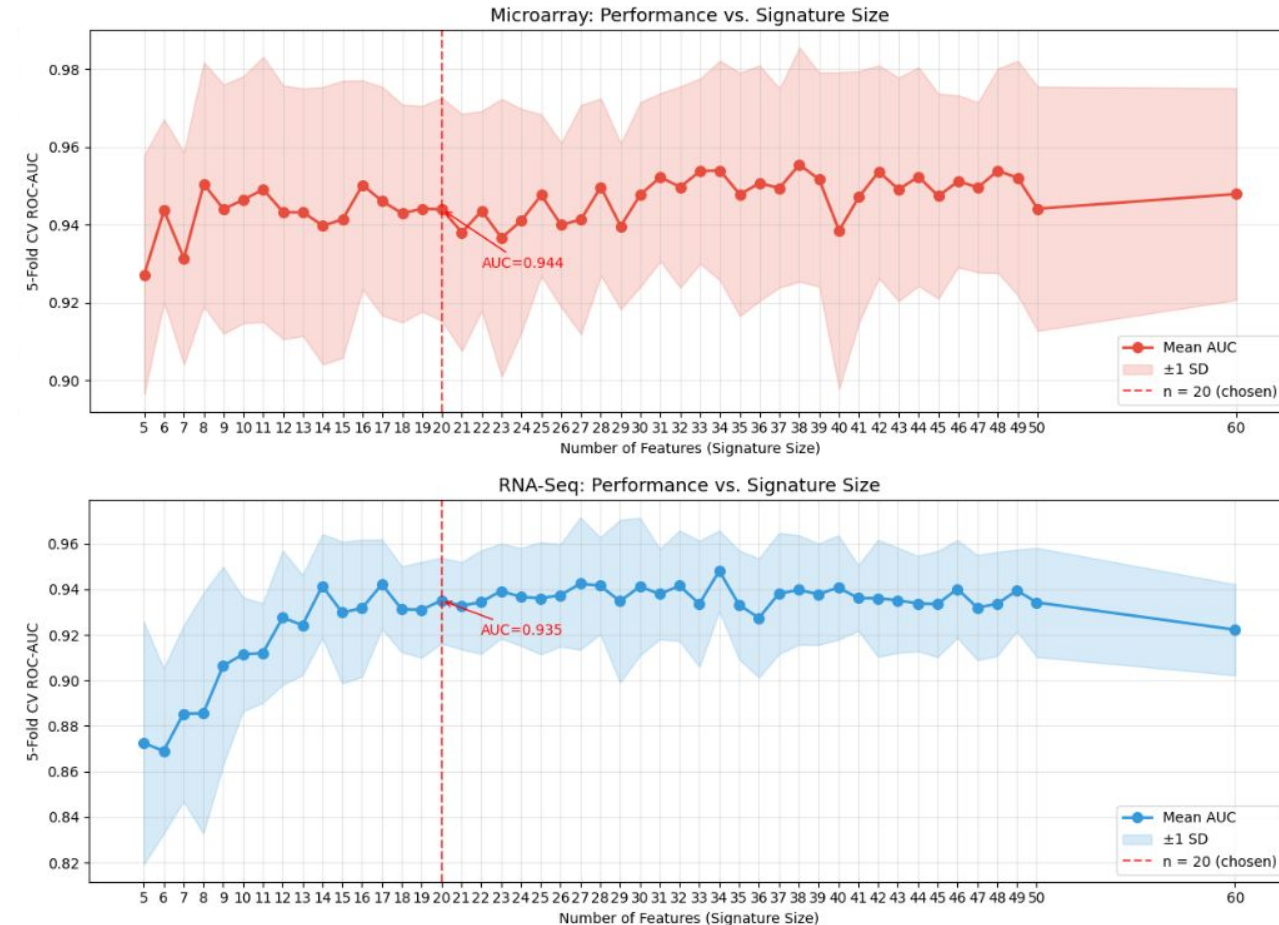
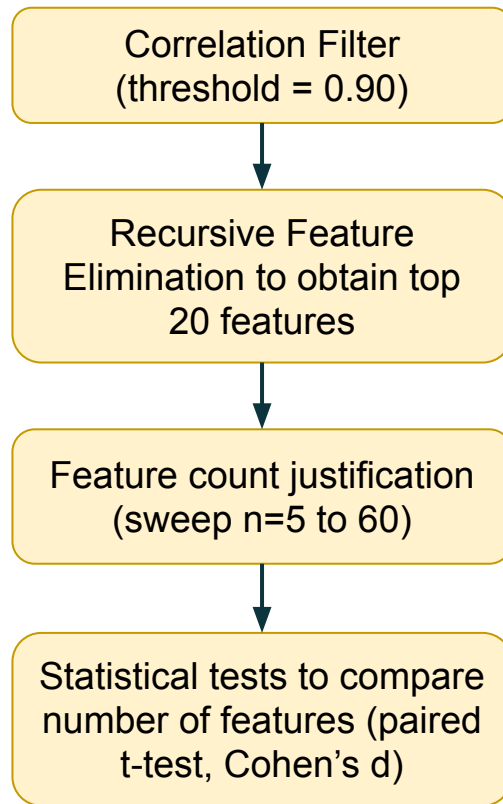
Supervised Machine Learning

Data Preprocessing And Feature Selection

Data Preprocessing

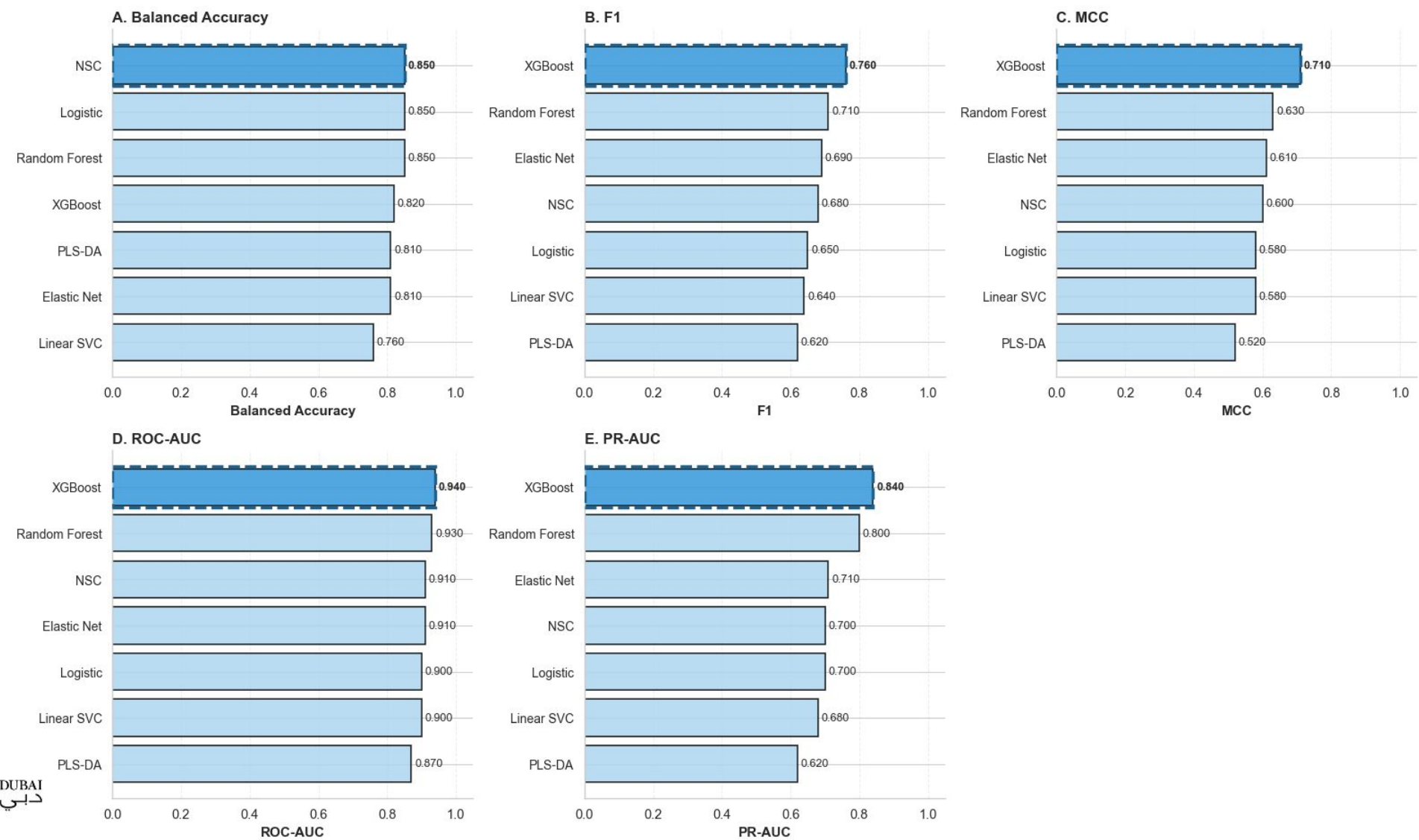


Feature Selection



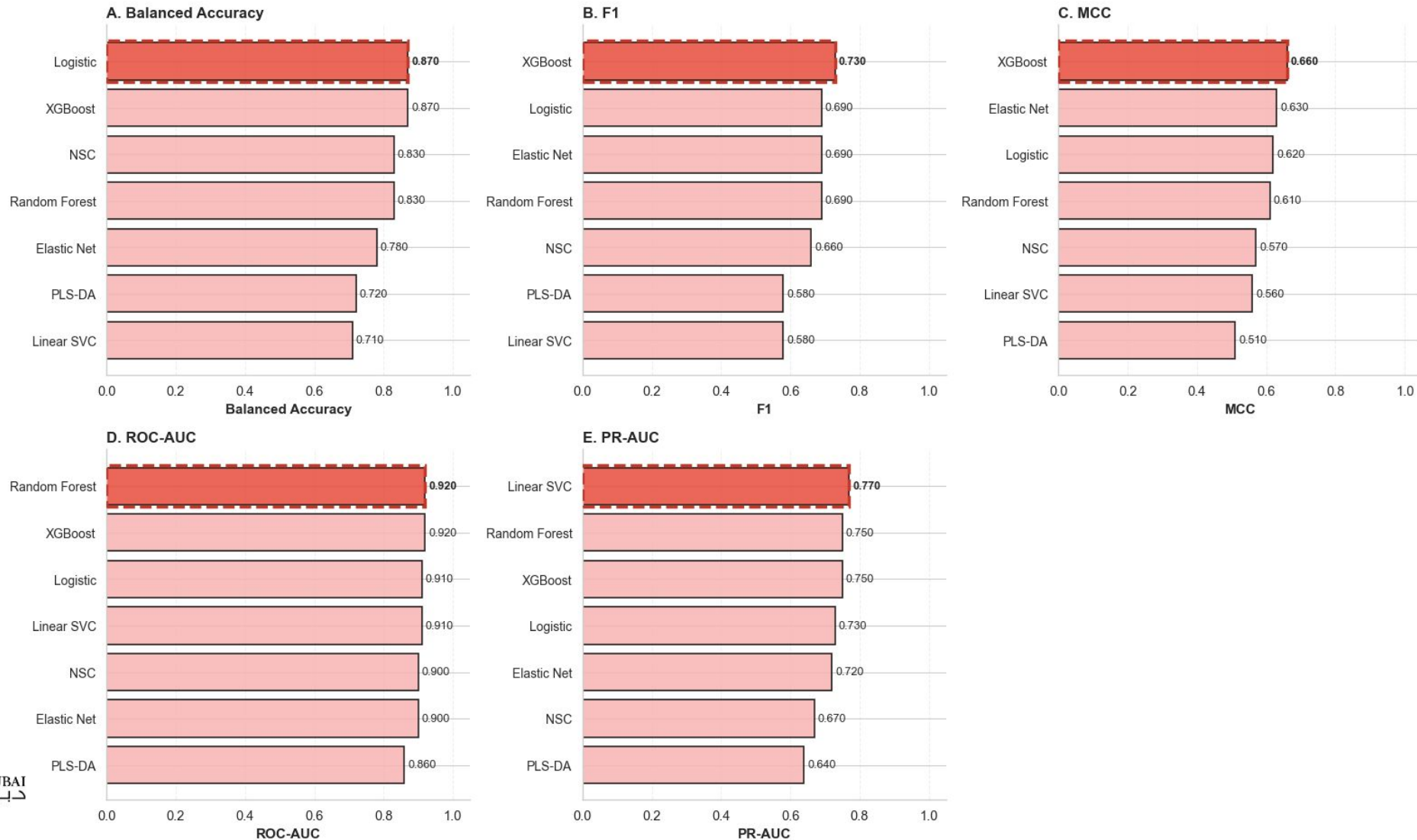
XGBoost Shows Superior Predictive Performance Across Evaluation Metrics & Platforms

RNA-seq Model Performance (Sorted by Performance)

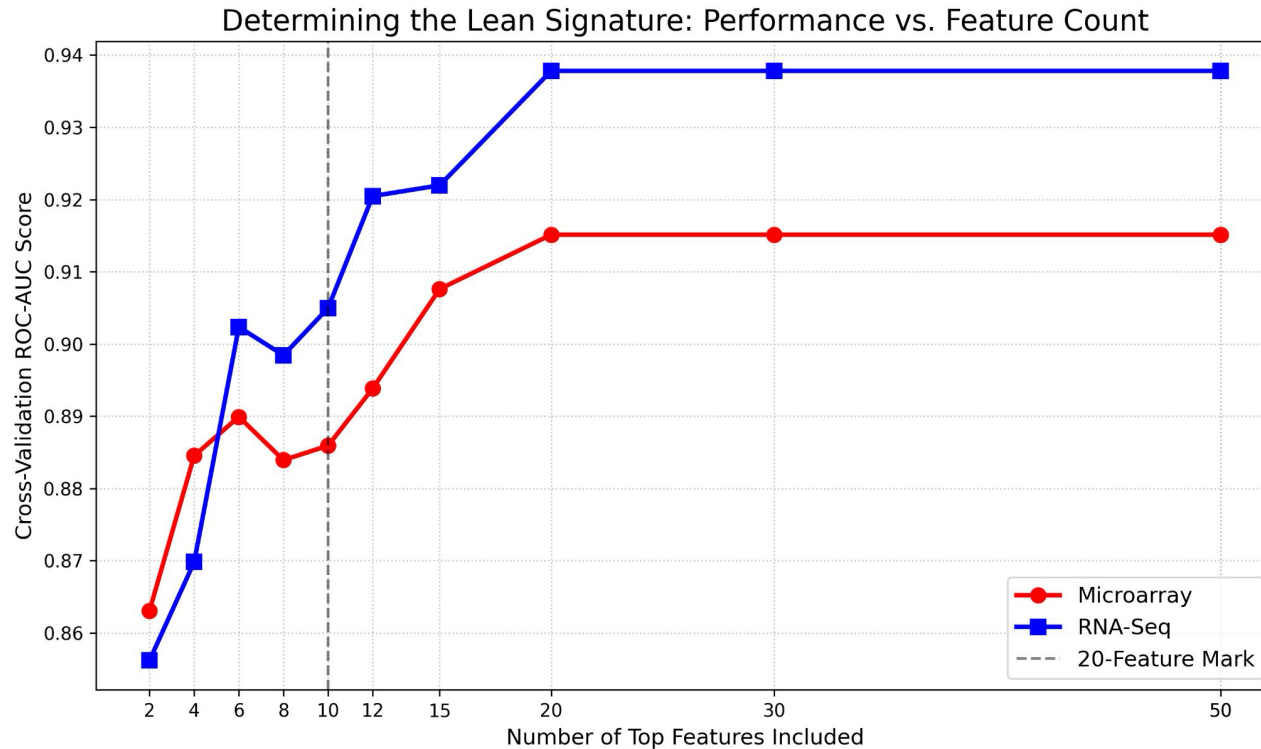


XGBoost Shows Superior Predictive Performance Across Evaluation Metrics & Platforms

Microarray Model Performance (Sorted by Performance)



Model Efficiency: The "Lean" Signature

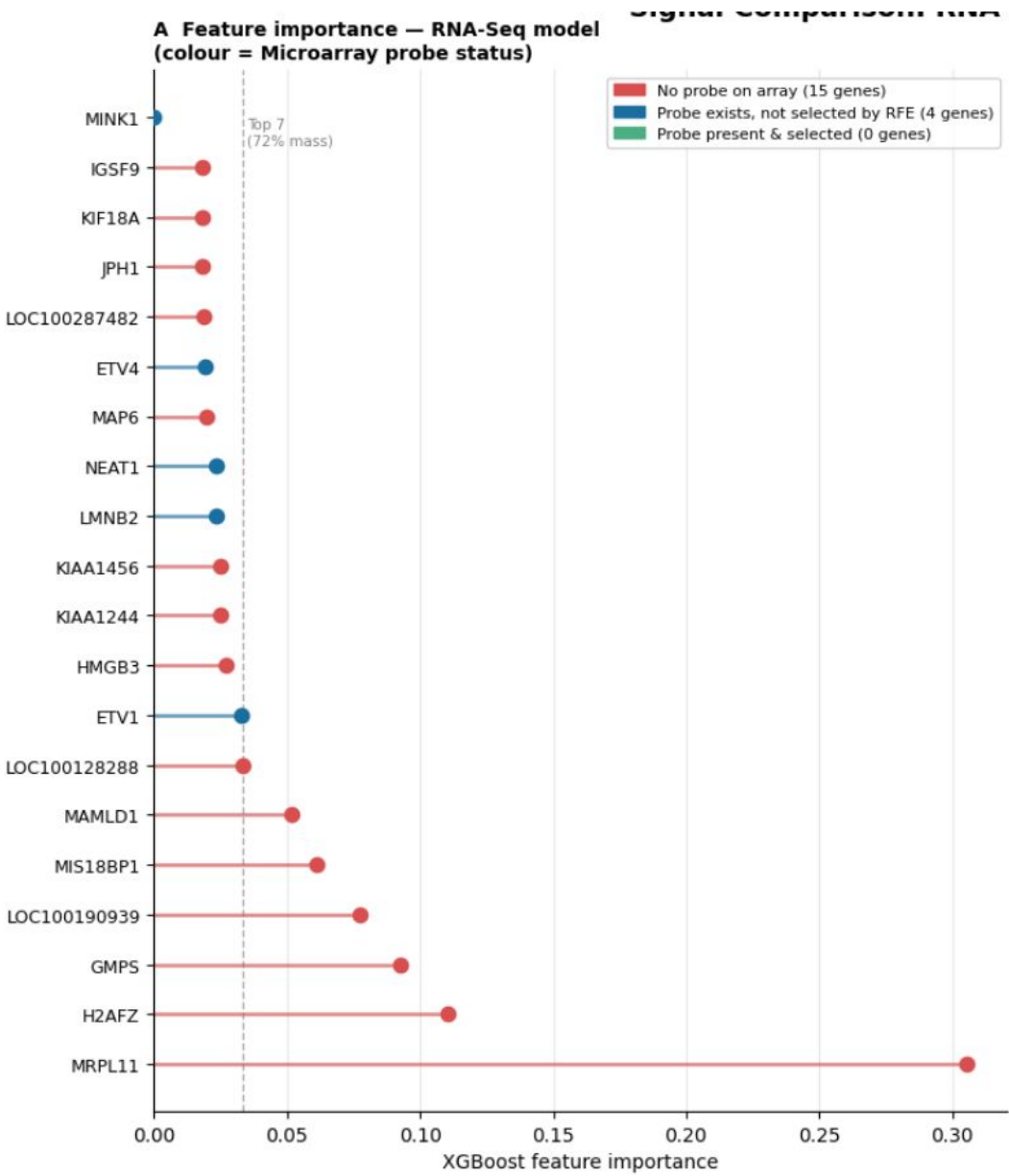


Metric	Microarray(10 Features)	RNA-Seq(10 Features)
Accuracy	0.8353	0.8394
Balanced Accuracy	0.8018	0.8553
Sensitivity (Recall)	0.7451	0.8824
Specificity	0.8586	0.8283
F1-Score	0.6496	0.6923
MCC	0.552	0.6162
ROC-AUC	0.8859	0.905

- RNA-Seq demonstrates higher predictive efficiency.
- When restricted to 10-features, RNA-Seq maintains a lead.



Resolution: Does Detail Matter?



Platform	Balanced Accuracy	Recall	Specificity	F1	MCC	ROC-AUC	PR-AUC
Microarray	0.873	0.8824	0.8636	0.7317	0.664	0.9191	0.7487
RNA-Seq	0.8207	0.6667	0.9747	0.7556	0.7122	0.9368	0.836

15 of 20 top RNA-Seq transcripts have no Microarray probe- most of the total model importance from RNA-seq is invisible to Microarray

All 5 genes present on both platforms show weaker signal in Microarray and were not selected by RFE.

RNA-Seq finds predictive transcripts that the Microarray was never designed to see. A targeted panel built from this signature could bridge the gap at deployment

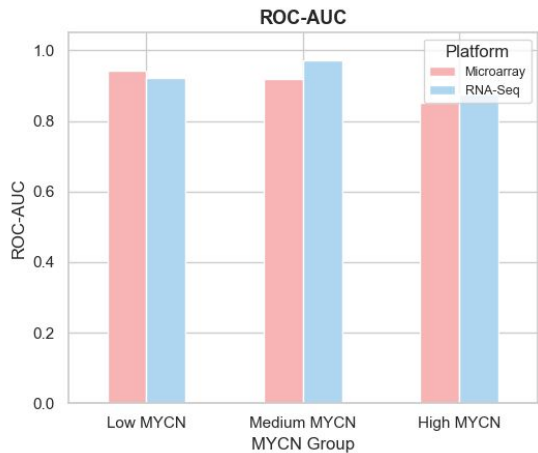
Detail matters for biomarker discovery, but the decision to incorporate RNA-seq depends on clinical use-case.
A leaner microarray model might be sufficient for routine clinical deployment.

Clinical Value: Beyond the Basics

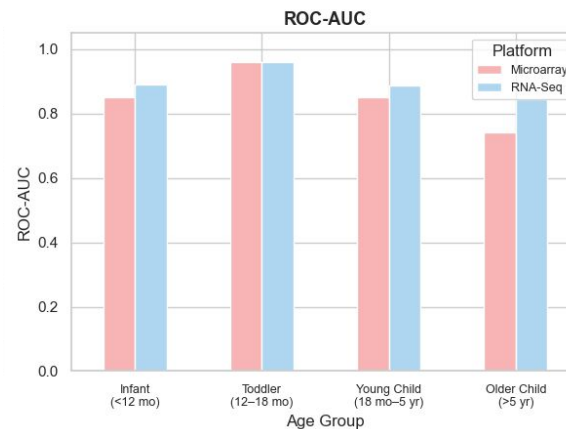
Impact of MYCN and Age on the analysis

- MYCN and age not part of selected features

MYCN expression



Age groups

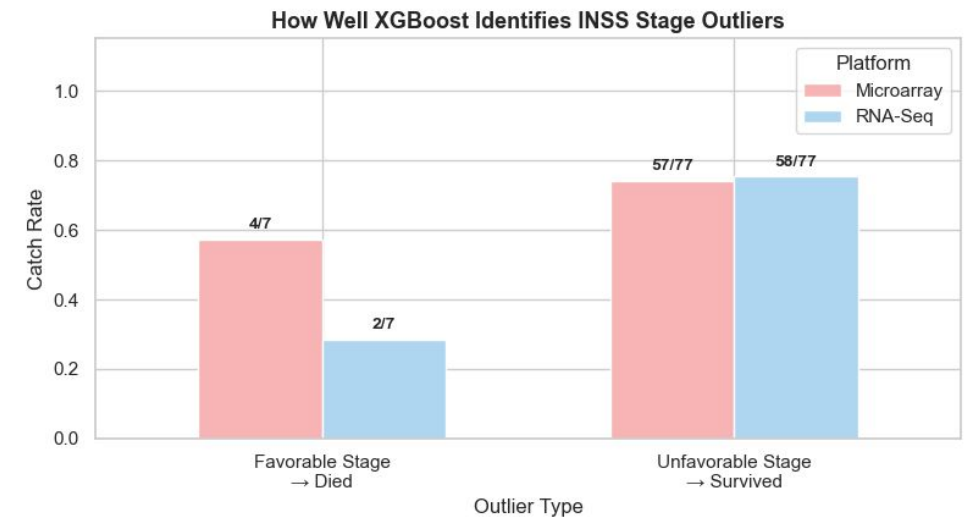


Model predicts the outcome without MYCN or age



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Identifying cases INSS staging missed



- favorable stages: 1, 2, 4S
- unfavorable stages: 3, 4

Microarray model predicts 4/7 “unfavorable” patients

Discussion

Limitations in the data:

- Class imbalance (51 vs 198) limits sensitivity even after including a weight factor for correction
- Mild overfitting in both models (~ 0.04 train-CV gap): regularization helps but doesn't fully eliminate it
- Test labels are unavailable, so all metrics are CV-based on training data: true test performance is unknown
- Only 5/20 RNA-Seq top genes had any Microarray probe, making the cross-platform gene-level comparison limited in scope
- RFE step relies on Random Forest rankings, which have randomness: results may shift slightly based on seed

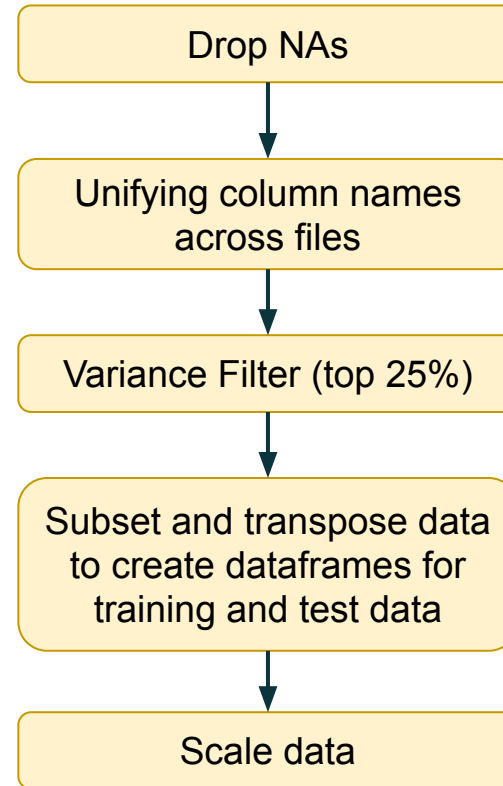
How this work can be extended in the future:

- Validate on an external neuroblastoma cohort
- Incorporate clinical covariates (MYCN, age) into a hybrid model
- Apply deep learning on raw transcript counts to capture splicing patterns
- Use SHAP values for more interpretable feature importance
- Investigate biological function of MRPL11, GMPS, H2AFZ in neuroblastoma prognosis

Questions

What are the key data pre-processing steps you have taken? How did they impact the result?

Data Preprocessing



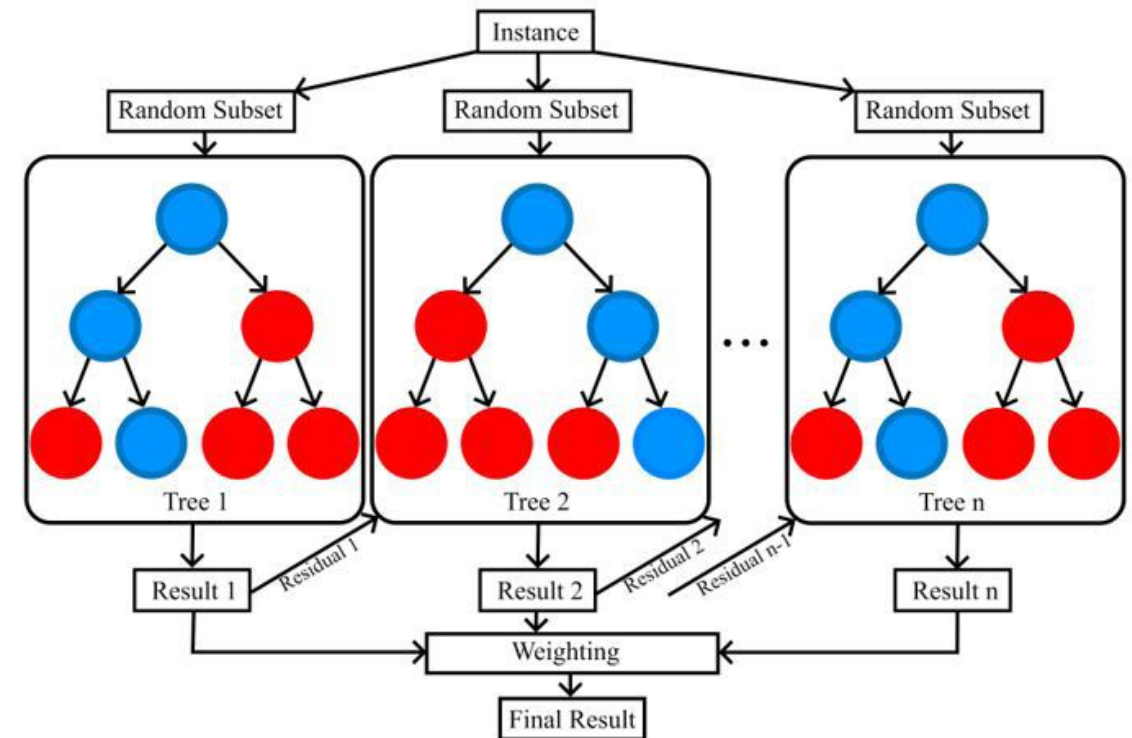
What were your rationale for choosing these algorithms (Both Supervised and Unsupervised)?

Data Context

- **Binary classification task** (death vs survival) with labeled data
- **High-dimensional biomedical data with complex biology**
→ non-linear relationships and feature interactions expected
- **class imbalance** (fewer death cases)

Model Choice

- **Linear models (e.g. Logistic Regression)**
→ assume linear relationships and may miss complex patterns
- **Tree-based ensembles (RF, XGBoost)**
→ capture non-linearity and interactions automatically
→ robust to noisy, high-dimensional data
- **XGBoost**
→ boosting focuses on misclassified (hard) samples
→ improves minority class prediction and overall performance



Did you run any un-supervised analysis? Do you notice good concordance between your observations from unsupervised analysis and supervised analysis?

Performed PCA on RNA-seq expression data (top variable genes)

Visualized samples by:

- Death from disease (binary)
- INSS stage (multiclass)

No clear clustering observed for either outcome

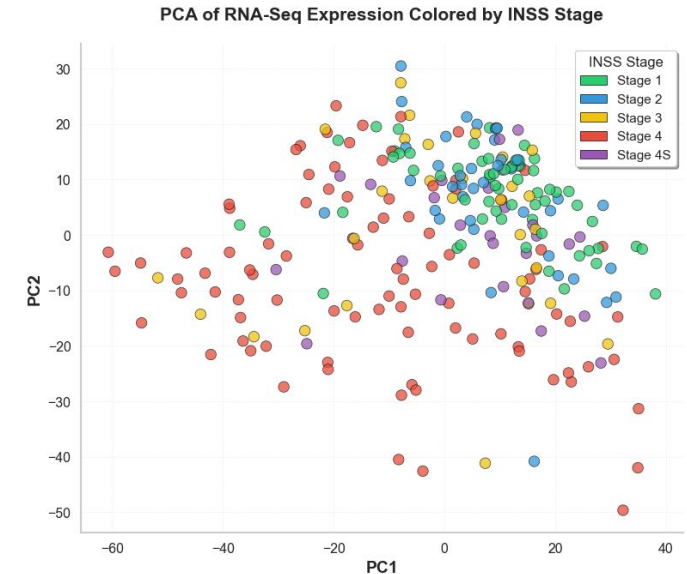
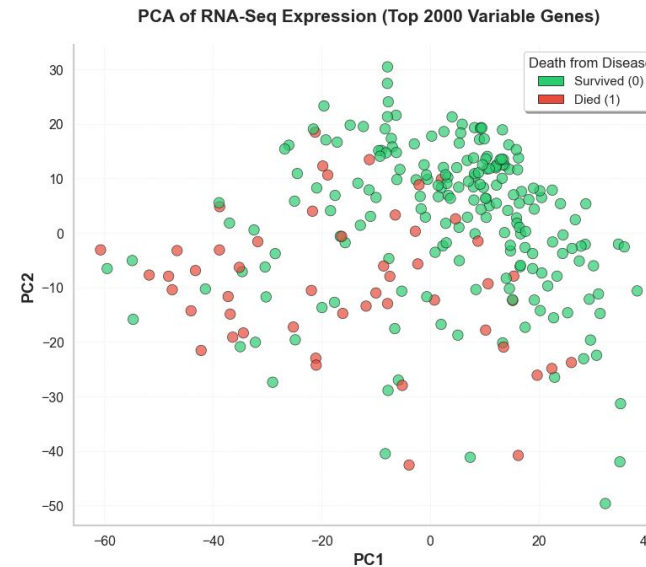
Significant overlap between classes → weak linear separability

Concordance with supervised results:

- Binary prediction performed well (higher MCC, F1)
- Multiclass (INSS stage) showed lower performance

Interpretation:

- Data exhibits complex, overlapping structure
- Models can extract subtle predictive signals
- Fine-grained classification remains challenging



What are the key findings? How do you ensure your results are reproducible?

Key Findings

- **RNA-based model shows highest predictive performance**, even without clinical data
- **Microarray-based lean model is sufficient for prediction**
 - detailed features more relevant for **biomarker discovery** than pure prediction
 - performance doesn't improve by that much to warrant the extra cost
- **Model identifies high-risk patients beyond INSS staging**
 - suggests added predictive value over clinical staging

Reproducibility

- **Reproducible pipeline**
 - fixed random seed and fully documented code
- **Cross-validation framework** to evaluate performance cross folds
- **Overfitting assessment**
 - small gap between train and CV performance
 - low standard deviation across CV folds

Test set predictions

RNA-Seq	
Dead	Alive
36	213

Microarray	
Dead	Alive
77	172

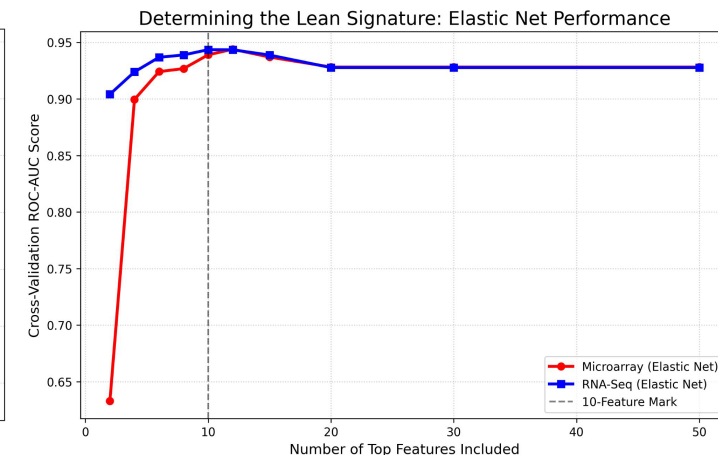
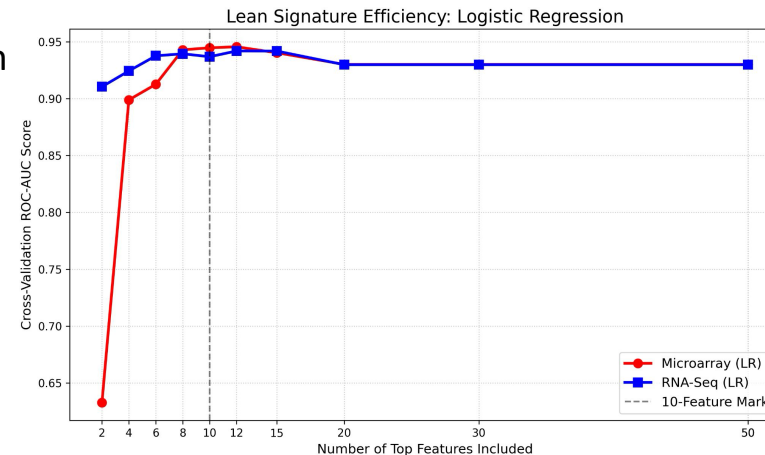
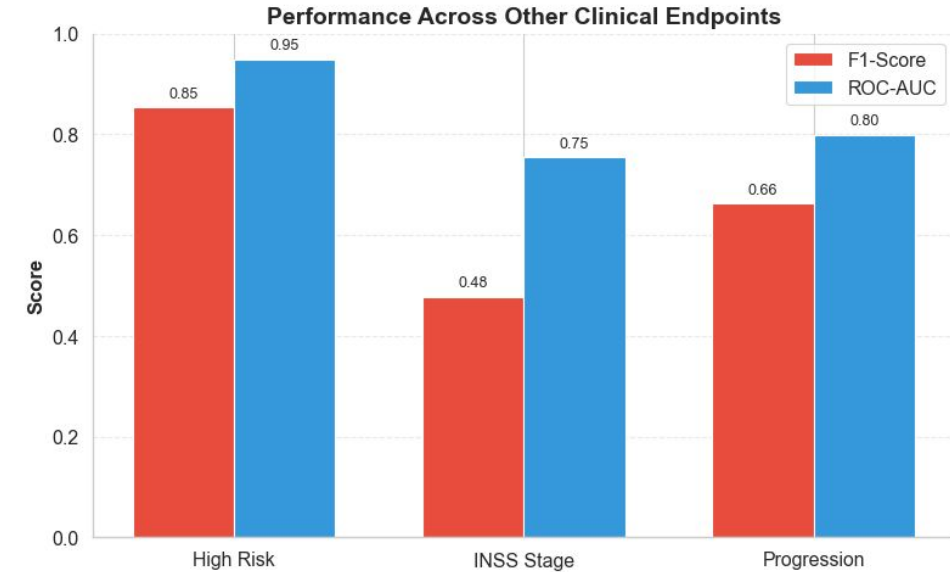
What are the key limitations of your analysis? How can you improve?

Limitations

- Performance decreases for secondary endpoints
→ Indicates reduced generalizability beyond primary outcome
- Multiclass prediction (INSS stage) remains challenging
→ Class imbalance and overlapping feature space
- Feature selection may introduce bias
→ RFE based on tree models may favour XGBoost over linear models

Improvements

- Explore model-specific or ensemble feature selection methods
- Apply techniques to address class imbalance (e.g. resampling, weighting)
- Incorporate biologically informed feature selection
- Validate findings on external datasets



Thank you for your time

